Alok Ranjan

Contact Information

M.Tech
Department of Aerospace Engineering
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Education

Year	Degree	Institute/School
2024-Present	**M.Tech Aerospace Engineering**	**Indian Institute of Technology, Kanpur**
2018-2022	**B.Tech Aerospace Engineering**	**Indian Institute of Space Science and Technology, Thiruvananthapuram**
2017	Senior Secondary	Prayas Boys Residential School, Raipur
2015	Secondary	Loyola Higher Secondary School, Kunkuri

Academic Achievements

- Selected for B.Tech in Aerospace Engineering at **Indian Institute of Space Science and Technology** (IIST), Thiruvananthapuram through **JEE Advanced (2018).**
- Qualified **GATE 2024 (Aerospace)** and selected for M.Tech Propulsion at **IIT Kanpur .**

Research Projects & Experience

M.Tech Thesis: Numerical Investigation of Electrostatic Spray Dynamics

Guide: *Dr. Ashoke De, IIT Kanpur* *(July'25 - Present)*

- Developed **sprayElectroFoam**, a custom **OpenFOAM** solver implementing EHD coupling, Lagrangian tracking, and image-charge forces.
- Implemented **RANS and LES turbulence models** on HPC to analyze transient spray plume dynamics and droplet stability.
- Post-processed high-density simulation datasets and validated droplet morphology against experimental benchmarks using **Python**.
- Managed end-to-end CFD workflows, covering geometry preparation, **meshing**, parallel solver execution, and statistical data extraction.

B.Tech Thesis: Failure Analysis of Bio-Inspired Self-Similar Hierarchical Composites

Guide: *Dr. Anup S, IIST Thiruvananthapuram* *(Jan'22 - May'22)*

- Performed **FE simulations in ABAQUS** to study **strain fields and stress distribution**.
- Computed **stiffness** using **computational and analytical methods**.
- Designed and fabricated **1H and 2H hierarchical composites via 3D printing** and conducted **tensile tests**.

Summer Internship: CFD Analysis of 2D Flow past Square Cylinder *(June'21 - Oct'21)*

Guide: *Dr. Manu K V, IIST Thiruvananthapuram*

- Simulated **unsteady flow and vortex shedding** at low Reynolds numbers using **OpenFOAM**.
- Computed **lift/drag coefficients and Strouhal numbers**, validated with benchmark data.

Course Project: Aerospace Vehicle Design *(Aug'21 - Nov'21)*

Mentored by: *Dr. Ayyappan G, VSSC, ISRO and Dr. Manoj T Nair, IIST, ISRO*

- Designed and optimized **IAD geometry, parachutes**, Developed codes for trajectory analysis.

Course Project: Parallel Computation for Sparse Matrix System *(Jan'25 - May'25)*

Instructor: *Dr. Ashoke De and Dr. Malay K Das, IIT Kanpur*

- Implemented **serial, OpenMP, and MPI solvers** for sparse matrices and **benchmarked performance and scalability** using **PETSc**.

Key Skilled Areas

- **Computational Fluid Dynamics**
- **Parallel Computation**
- **Fluid Dynamics**
- **Numerical Linear Algebra & Iterative Solvers**

Technical Skills

• **OpenFOAM** • **PETSc** • **OpenMPI** • **HPC** • **Python** • **SolidWorks, CATIA, Inventor, Abaqus** • **LaTeX, ParaView, ICEM CFD, Gmsh,**

Relevant coursework

M.Tech: **Parallel Computation for Sparse Matrix System, Numerical Linear Algebra, Computational Methods, Fluid Dynamics,** Combustion, Heat Transfer, Compressible Flow.
B.Tech: Air-Breathing Propulsion, Thermodynamics, Fluid Mechanics, Heat Transfer, Industrial Engineering,Principles of Management Systems, Optimization Techniques in engineering.